

SAYAK SATPATHI

AI/ML Engineer

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AI & ML student who builds and **deploys** machine-learning and computer-vision systems end to end – an award-winning aerial-detection app, an edge CCTV pipeline, and a deployed fraud-detection app.

Education

Sikkim Manipal Institute of Technology (SMIT)

B.Tech, Computer Science & Engineering (AI & ML); Minor in Cybersecurity

GPA: 7.69/10 (through Semester 4) · Class XII (Science): 96.2%

Majitar, Sikkim

2024 – 2028

Technical Skills

Languages	Python, C++, Java, JavaScript, SQL
ML & CV	PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv8, CNNs/RNNs, NLP
Backend & Deploy	FastAPI, Flask, Streamlit, REST APIs, Docker
Data & Cloud	Pandas, NumPy, MySQL, MongoDB, AWS, Google Cloud
Tools	Git, GitHub, Linux, Data Structures & Algorithms, OOP

Projects

AeroMinds – Aerial Waste Intelligence (GitHub)

2026

Python, PyTorch, YOLOv8, CUDA, Streamlit, Docker

- Won ELCIA Next-Gen Hackathon (2nd Runner-Up of 180+ teams) with a fine-tuned YOLOv8 app that detects illegal dumping sites from drone footage and auto-scores severity (HIGH/MED/LOW) via cluster and coverage analysis.
- Built a real-time PyTorch + CUDA inference pipeline and Streamlit command center with an automated incident workflow (DETECTED → ASSIGNED → CLEARED); containerized with Docker for Hugging Face Spaces deployment.

Prahari – Edge AI Video-Intelligence for CCTV (GitHub)

2026

Python, OpenCV, Object Detection, Tracking, ANPR, FastAPI

- Built an edge-first CV layer adding real-time detection, tracking, and ANPR to existing CCTV, running offline and syncing on reconnect; developed for Smart India Hackathon 2026 (Ministry of Home Affairs / SSB).
- Engineered automatic ground-plane self-calibration accurate to 0.4–3.2% across a 5-camera fleet, and an alert-rationing governor that cut alerts from 1064 to 138 per 90s window while retaining 81% of events.

CyberShield AI (Live · GitHub)

2024

Python, Scikit-learn, Random Forest, Gradient Boosting, SVM, Streamlit

- Deployed a fraud-detection web app on Streamlit Cloud running a 5-model scikit-learn ensemble (Random Forest, Gradient Boosting, SVM, Isolation Forest, Logistic Regression) over uploaded transaction datasets, with an interactive analytics dashboard.

Smart Traffic Management System (Ongoing)

2025

Python, YOLOv8, OpenCV, Streamlit

- Built real-time vehicle and pedestrian detection with YOLOv8, using lane-wise density estimation to drive adaptive signal control on a Streamlit analytics dashboard.

Experience

HZ IT Company – Web Developer Intern

Jun 2026 – Jul 2026

- Built an IT service-request and contact-management portal with an admin dashboard, backed by a Node.js/Express REST API with full CRUD and a JavaScript/CSS client – delivered end to end.

Google Developer Groups (GDG) Kolkata – Open Source Contributor

Apr 2025 – Present

- Merged pull requests across 3+ repositories as an Open Source Connect 2026 contributor, delivering feature enhancements and bug fixes.

Hackathons & Achievements

- **ELCIA Next-Gen Hackathon** – 2nd Runner-Up, top 3 of 180+ teams (2026), team AeroMinds.
- **Smart India Hackathon 2026** – qualified Institute Round at Rank 7 with Project Prahari; advancing.
- **Adobe University Hackathon** (Unstop, 2026) – cleared Round 1.